25901118 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCBOT202 - INTRODUCTION TO FLOWERING PLANTS AND THEIR ECONOMIC IMPORTANCE

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Answer Key

Part A

Short answer Answer

Any **10** questions

Each question carries **1** mark

1. What is the significance of floral formula? [K] / [CO1]

Ans. Using letters, numbers, and signs it provides a systematic information of a flower's morphology for identification

2. What is the morphology of a flower bud? [K] / [CO1]

Ans. A vegetative bud transformed into a condensed, undeveloped reproductive shoot consisting of a short axis (thalamus) bearing closely packed floral organs (sepals, petals, stamens, and carpels).

3. What is aestivation? [K] / [CO1]

Ans. Aestivation is the arrangement of sepals or petals in a floral bud with respect to one another in the same whorl.

4. How many parts does a scientific name contain? [U] / [CO2]

Ans. A scientific name contains two parts (binomial). Generic epithet and specific epithet.

5. Name the three major divisions in Bentham & Hooker's system of classification. [U] / [CO2]

Ans. Dicotyledons , Gymnosperms and Monocotyledons

6. What is fumigation in herbarium maintenance? [K] / [CO2]

Ans. The process of treating herbarium specimens with chemicals or fumes to protect them from insects and fungal damage

7. Name the type of gynoecium found in Apocynaceae. [K] / [CO3]

Ans. Bicarpellary and apocarpous.

8. Name two economically important plants of the Caesalpiniaceae family. [K] / [CO3]

Ans. Any two plants of the family. (Cassia, Tamarindus…)

9. Identify the genus of rubber plant. [K] / [CO3]

Ans. **Hevea**

10. Identify the vegetable rich in iron and vitamins [K] / [CO3]

Ans. Amaranthus/Moringa

11. Which part of the Alpinia plant is mainly used for medicinal purposes? [K] / [CO4]

Ans. Rhizome

12. Name one plant used for shelter and house construction. [K] / [CO4]

Ans. Bambusa

[1x10 = 10]

Part B

Short Essay Answer

any **6** questions

Each question carries **5** marks

13. Give an account on dehiscent and indehiscent fruits. [U] / [CO1]

Ans. Dehiscent - **Legume:** splits along two sutures (e.g., pea)

Follicle: splits along one suture (e.g., *Calotropis*)

Capsule: splits in various ways (e.g., cotton)

Siliqua/Silicula: characteristic of mustard family

Indehiscent - **Achene:** single-seeded, free pericarp (e.g., sunflower)

Caryopsis: seed coat fused with pericarp (e.g., wheat)

Nut: hard, woody pericarp (e.g., cashew)

Drupe: fleshy outer part with hard stone (e.g., mango, coconut)

14. What is sorosis? Explain with suitable example? [U] / [CO1]

Ans. Sorosis is a type of **multiple fruit** that develops from a **spike or spadix inflorescence**, where the **entire inflorescence (flowers, axis, and sometimes bracts) fuses together** to form a single, large fruit.

Ananas comosus (Pineapple) –the axis and flowers fuse into a juicy mass.

Artocarpus heterophyllus (Jackfruit) – develops from a spike.

Rachis (axis): Thick, fibrous central core.

Edible portion: Consists of fleshy yellow bulbs. Each bulb develops from the perianth of individual flowers.

Seeds: Each bulb encloses a single large seed.

Fibrous strands: Non-edible, thread-like structures derived from unfertilized flowers.

Morus (Mulberry) – formed from a catkin inflorescence.

15. Explain the natural system of classification with reference to Bentham & Hooker [U] / [CO2]

Ans. classification groups plants based on their overall morphological similarities and natural affinities.

Based on observable characters like floral structure, vegetative traits, etc.

Reflects natural relationships among plants (not evolutionary).

Practical and widely used in herbaria.

Helps in easy identification

Evolutionary relationships not accounted**.**

16. State any five major identifying characters of the family Rubiaceae. [K] / [CO3]

Ans. **Leaves** opposite, simple, with **interpetiolar stipules**.

Inflorescence usually cymose.

Flowers bisexual, actinomorphic, epigynous.

Corolla gamopetalous (fused petals).

Ovary inferior, usually bicarpellary.

17. Discuss the nature of the staminal tube in Malvaceae. [U] / [CO3]

Ans.Stamens are **numerous** and their **filaments are fused** to form a **staminal tube (column)**.

This condition is called **Monadelphous**.

The tube **surrounds the style**, and **anthers remain free** at the top.

It is a **distinctive diagnostic feature** of Malvaceae.

18. Write a short note on Neem as a medicinal plant. [K] / [CO3]

Ans. Leaves: **antibacterial, antifungal**, used in skin diseases.

Bark: used for **fever and malaria**.

Seeds: yield **neem oil**, used in treating infections.

Twigs: used as **natural toothbrush** (oral hygiene).

19. Examine the food and non-food uses of green gram. [An] / [CO3]

Ans. **Food uses:**

- Consumed as dal, curry, sprouts.
- Rich in protein, vitamins, and minerals.
- Easily digestible; used in infant and patient diets.

Non-food uses:

- Used as green manure to improve soil fertility.
- Fodder for livestock.
- Helps in nitrogen fixation (leguminous crop).

20. Explain how ethnobotany helps in tribal welfare and development. [U] / [CO4]

Ans. Healthcare: Identifies traditional herbal medicines for primary healthcare.

Livelihood: Promotes income through medicinal plants, handicrafts, forest products.

Conservation: Preserves traditional knowledge and biodiversity.

Sustainable use: Encourages eco-friendly utilization of plant resources.

Cultural preservation: Maintains traditions and indigenous practices. **[5x6 = 30]**

Part C

Essay

Answer any **1** questions Each

question carries **10** marks

21. What is a fruit? Give an account on simple fruits. [K] / [CO1]

Ans. A **fruit** is a **matured ovary** of a flower formed after fertilization, containing **seeds**. Sometimes, other floral parts may also contribute to fruit formation (accessory fruits).

Simple fruits develop from the **single ovary of a single flower**. They may be **dry or fleshy**, depending on the nature of the pericarp.

Dry Simple Fruits

Dehiscent fruits (split open at maturity)

Legume: Splits along two sutures (e.g., pea)

Follicle: Splits along one suture (e.g., *Calotropis*)

Capsule: Opens in different ways (e.g., cotton)

Siliqua: Long fruit of mustard family

Indehiscent fruits (do not split)

Split into **mericarp units**

Examples: coriander, castor

Fleshy Simple Fruits

Drupe -Outer fleshy part with **hard inner stone**

Example: mango, coconut

Berry - Entire pericarp fleshy

- Example: tomato, grape, banana

Pome- Fleshy thalamus forms edible part

- Example: apple, pea.

Pepo – Cucurbita.

22. With the help of a labelled diagram explain the inflorescence of Poaceae. [U] / [CO3]

Ans. The inflorescence in Poaceae is composed of spikelets, which are the basic units. Each spikelet consists of:

Glumes
- Two sterile bracts at the base of the spikelet.
- Protect the developing florets**.**

Rachilla
- The central axis of the spikelet on which florets are arranged.

Florets
- Each floret is a small flower enclosed by:
  - Lemma (outer bract, often with an awn)
  - Palea (inner bract)

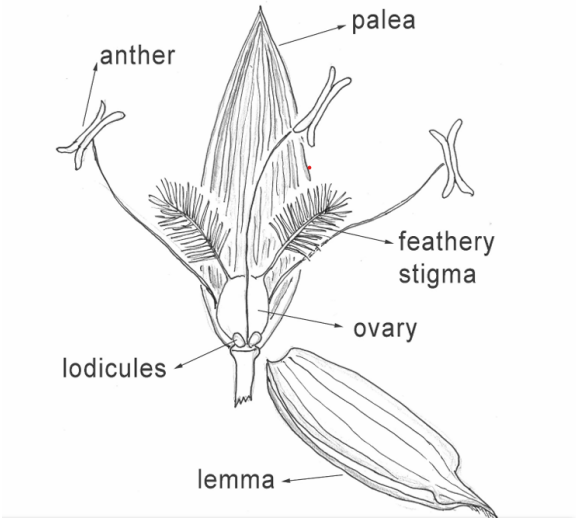
Lodicules
- Small scale-like structures (represent reduced perianth).
- Help in opening of the floret during flowering.

Androecium

- Usually 3 stamens with versatile anthers.

Gynoecium

- Superior ovary, unilocular with a single ovule.
- Feathery stigmas for wind pollination.



[10x1 = 10]